

Jules ZHU Haoyu

Sun Yat-sen University – 519082 Zhuhai, Guangdong – China

☎ (+86) 13302264395 • ✉ zhuhy86@mail2.sysu.edu.cn

🌐 <https://jules-zhu.github.io/> • Gender: Male

Education Background

Sun Yat-sen University

Undergraduate, School of Mathematics (Zhuhai)

Zhuhai

2022 – present

Gap because of Tuberculosis

April 2025 – Sept 2026

Expected date of graduation

June 2027

Courses

Commutative Algebra and Homological Algebra (Graduate)

Feb 2024 – Jun 2024

- **Textbook:** *Introduction to Commutative Algebras*, M.F. Atiyah, I.G. Macdonald, and other materials.
- **Instructor:** Prof. Lei Zhang.
- **Grade point:** 77/100.

Representation Theory (Graduate)

Feb 2024 – Jun 2024

- **Textbook:** *Introduction to Lie Algebras and Representation Theory*, J.E. Humphreys.
- **Instructor:** Prof. Min Huang.
- **Grade point:** 96/100.

Real Analysis (Graduate)

Sept 2024 – Jan 2025

- **Textbook:** *Measure Theory and Fine Properties of Functions*, L.C. Evans, R.F. Gariepy.
- **Instructor:** Prof. Xining Li.
- **Grade point:** 85/100.

Complex Analysis (Graduate)

Feb 2026 – Jun 2026

- **Textbook:** *A Course in Complex Analysis and Riemann Surfaces*, Wilhelm Schlag.
- **Instructor:** Prof. Hang Xu.
- **Grade point:** TBA.

Modern Number Theory (Graduate)

Feb 2026 – Jun 2026

- **Textbook:** *A Classical Introduction to Modern Number Theory*, Kenneth Ireland, Michael Rosen.
- **Instructor:** Prof. Chan leong Kuan.
- **Grade point:** TBA.

Algebraic Geometry 2 (Seminar)

Feb 2026 – Jun 2026

- **Textbook:** *Algebraic Geometry*, Robin Hartshorne.
- **Speakers:** Xinyu Fang, Kang Huang, Boping Jiang, Haoyu Zhu.

Research Experience

Sun Yat-sen University

Zhuhai, Guangdong

College Students' Innovative Training Plan Program

Oct. 2023 – Oct. 2024

- *Summary:* Read Stembridge's paper on Immanants and worked on problems involving integer partitions and irreducible characters of symmetric groups. This research is under the supervision of Prof. Min Huang (Sun Yat-sen University).

Peking University

Beijing

PKU Algebraic Combinatorics Experience

Jul. 2024 – Aug. 2024

- *Summary:* On Bott–Samelson rings for Coxeter groups. One can see [arxiv.org/abs/2408.10155] for details. This research lasts for 8 weeks under the supervision of Dr. Tao Gui (Peking University) and in collaboration with Lin Sun and Shihao Wang.

Sun Yat-sen University

Zhuhai, Guangdong

College Students' Innovative Training Plan Program

Oct. 2024 – Oct. 2025

- *Summary:* Comparing skein algebras and quantum cluster algebras of certain semisimple Lie algebras. This research is supervised by Prof. Min Huang (Sun Yat-sen University).

The Hong-kong University of Science and Technology

Hongkong

2025 Summer Research Program (on campus)

Jul. 2025 – Aug. 2025

- *Summary:* Studying whether $\mathcal{A} = \mathcal{U}$ for cluster algebras of type G_2 from moduli spaces of G -local systems. This research is supervised by Prof. Ivan Chi-Ho Ip (HKUST).

Skills and Miscellaneous

Programming: MATLAB, Python, C++, SageMath

Tools: LaTeX, Markdown

Languages: Mandarin (Native), English (Equivalent C1), French (DELFB2, 2026 in Canton: 61/100)

Hobbies: Piano (ABRSM Grade 8), Chorus, Football, Movie

Honors and Awards

2023: Third Prize Scholarship.

2024: Excellent Student Cadre.